

1 Introduction

$$|\Psi\rangle = \sum_{i_1=1}^{d_1} \sum_{i_2=1}^{d_2} \cdots \sum_{i_N=1}^{d_N} \Psi_{i_1,i_2,\dots,i_N} |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle \quad (1)$$

$$T \in \mathbb{C}^{\chi_1 \times \chi_2 \times \cdots \times \chi_n} \quad (2)$$

$$T_{i_1,i_2,\dots,i_n} \in \mathbb{C}, \quad i_j \in \{1,2,\dots,\chi_j\} \quad (3)$$

$$c = \sum_{\alpha=1}^{\chi} A_{\alpha} B_{\alpha} \quad (4)$$

$$C_{ij} = \sum_{\alpha=1}^{\chi_2} A_{i\alpha} B_{\alpha j} \quad (5)$$

$$\Psi_{i_1,i_2,\dots,i_N} := \sum_{\alpha_1=1}^{\chi_1} \sum_{\alpha_2=1}^{\chi_2} \cdots \sum_{\alpha_{N-1}=1}^{\chi_{N-1}} T_{1,\alpha_1}^{[1],i_1} T_{\alpha_1,\alpha_2}^{[2],i_2} \cdots T_{\alpha_{N-1},1}^{[N],i_N} \quad (6)$$

$$d^N \rightarrow N\chi^2 d \quad (7)$$

$$W \in \mathbb{C}^{n \times m}, \quad n \geq m \quad (8)$$

$$W^\dagger W = \mathbb{1} \quad (9)$$

$$WW^\dagger = \mathbb{P}, \quad \mathbb{P}^2 = \mathbb{P} \quad (10)$$

$$U \in \mathbb{C}^{n \times n} \quad (11)$$

$$U^\dagger U = \mathbb{1} \quad (12)$$

$$UU^\dagger = \mathbb{1} \quad (13)$$

$$\langle \Psi | \hat{O}_n | \Psi \rangle = \quad (14)$$

$$|\Psi(\Delta t)\rangle = \hat{U}(\Delta t) |\Psi\rangle = e^{-i\Delta t \hat{H}} |\Psi\rangle \quad (15)$$

$$\hat{H} = \sum_{j \text{ even}} \hat{h}^{[j,j+1]} + \sum_{j \text{ odd}} \hat{h}^{[j,j+1]} =: \hat{H}_{\text{even}} + \hat{H}_{\text{odd}} \quad (16)$$

$$\hat{U}(\Delta t) = e^{-i\Delta t(\hat{H}_{\text{even}} + \hat{H}_{\text{odd}})} = \underbrace{e^{-i\Delta t\hat{H}_{\text{even}}}e^{-i\Delta t\hat{H}_{\text{odd}}}}_{\hat{U}^{\text{TEBD1}}(\Delta t)} + \mathcal{O}(\Delta t^2) \quad (17)$$

$$e^{-i\Delta t\hat{H}_{\text{even}}} = e^{-i\Delta t \sum_{j \text{ even}} \hat{h}^{[j,j+1]}} = \prod_{j \text{ even}} e^{-i\Delta t \hat{h}^{[j,j+1]}} \quad (18)$$

$$\Psi_{i_1, i_2, \dots, i_N} := \mathcal{C} \left(T^{[1], i_1}, T^{[2], i_2}, \dots, T^{[N], i_N} \right) \quad (19)$$

$$\Lambda^{[n]} B^{[n+1]} \approx A^{[n]} \Lambda[n+1] \quad (20)$$

$$\Lambda^{[n]} \approx A^{[n]} \Lambda \quad (21)$$

$$\|\Lambda^{[n]} - A^{[n]} \Lambda\| \quad (22)$$

$$\hat{H} = \underbrace{\sum_{x=1}^{L_x} \hat{H}_x}_{\text{columns}} + \underbrace{\sum_{y=1}^{L_y} \hat{H}_y}_{\text{rows}} \quad (23)$$

$$\| |\Psi\rangle - |\Psi'\rangle \| = \sqrt{\langle \Psi | \Psi \rangle + \langle \Psi' | \Psi' \rangle - 2 \langle \Psi' | \Psi \rangle} \quad (24)$$

$$= \sqrt{2 - 2 \operatorname{Re} \langle \Psi' | \Psi \rangle} \quad (25)$$

$$(T'_{\text{opt}}, W'_{1,\text{opt}}, W'_{2,\text{opt}}) = \operatorname{argmax}_{T, W'_1, W'_2} \operatorname{Re} \langle \Psi' | \Psi \rangle \quad (26)$$

$$T'^{\dagger} T' = \mathbb{1}, \quad W_1'^{\dagger} W_1' = \mathbb{1}, \quad \|W_2'\|_{\text{F}} = 1 \quad (27)$$

$$f_{\text{trunc}}(U, \theta) = \sqrt{\sum_{\mu=\chi+1}^{\chi D^2} S_{\mu}^2} \quad (28)$$

$$f_{\text{Rényi}}(U, \theta, \alpha) = \frac{1}{1-\alpha} \log \operatorname{Tr}(\rho^{\alpha}) = \frac{1}{1-\alpha} \log \left(\sum_{\mu=1}^{\chi D^2} S_{\mu}^{2\alpha} \right) \quad (29)$$

$$\rho = \operatorname{Tr}_{(j,r)}(|\tilde{\theta}\rangle \langle \tilde{\theta}|) \quad (30)$$

$$\operatorname{Tr} \rho^2 \quad (31)$$

$$\mathcal{O}(N_{\text{iter}} D^9) \quad (32)$$

$$\mathcal{O}(N_{\text{iter}}N_{\text{tCG}}D^9) \quad (33)$$

$$\mathcal{O}(D^9) \rightarrow \mathcal{O}(D^8 + N_{\text{svd}}) \quad (34)$$

$$\mathcal{O}(D^9) \rightarrow \mathcal{O}(D^8) \quad (35)$$

$$|\Psi\rangle = \hat{U}^{[i,j]}(\Delta t) |\Psi\rangle \quad (36)$$

$$(T'_{i,\text{opt}}, T'_{j,\text{opt}}, W'_{1,\text{opt}}, W'_{2,\text{opt}}, W'_{3,\text{opt}}) = \quad (37)$$

$$\operatorname{argmax}_{T'_i, T'_j, W'_1, W'_2, W'_3} \operatorname{Re} \langle \Psi' | \hat{U}^{[i,j]}(\Delta t) | \Psi \rangle \quad (38)$$

$$\hat{U}(\Delta t) = \hat{U}^{\text{TEBD1}}(\Delta t) + \mathcal{O}(\Delta t^2) \quad (39)$$

$$\hat{U}(\Delta t) = \hat{U}^{\text{TEBD2}}(\Delta t) + \mathcal{O}(\Delta t^3) \quad (40)$$

$$g \approx 3.04438 \quad (41)$$

$$/, \quad (42)$$

$$/, \quad (43)$$